Mathematical Models and Physical Correspondence

Kirtus Justus (Theoretical Physicist)
Port Coquitlam, British Columbia, Canada
10 November 2026 / B.C. Pacific Time

1. The research question and the role of representations

The program asks what would justify interpreting a proposed relational, informational or algebraic structure as a physical model. This includes questions about possible primitives, but also concrete problems of identification: which properties are shared by every description compatible with specified preparations, interventions and observations? Progress can consist of a conditional theorem, a counterexample to a proposed implication, an identifiable parameter, or a useful prediction. It need not settle every ontological question simultaneously. [26, §§1, 3; 27, “Identification from observations”]

A mathematical information channel can be a map from inputs to outcome distributions without an assigned physical bandwidth, duration or energy. Similarly, a coordinate, spectral parameter, scale index or iteration count can organize a mathematical representation before acquiring a physical interpretation. Such representations are legitimate objects of study. Claims about hertz, elapsed time, energy expenditure or causal propagation require additional correspondence rules and experimental warrant; they are not supplied by the representation’s name. [1, §§2, 5, 8; 22, §4]

The signed-sector signal proposal is one exploratory direction. Its notation and proposed route toward effective geometry require specified carriers, relations, equivalences, composition and observation laws. Relational-core models, conventional quantum and classical trials, and analytic field or operator models provide distinct comparison settings. Their assumptions are not interchangeable, and their successes do not collectively establish one universal ontology. The references identify mathematical manuscripts and exact research-note passages. Claims available only as review-note statements are identified as reported assertions. This overview states scope and dependencies; it does not replace the individual proofs or independently certify every analytic chain. [1, §§3-4; 11, §§1-3; 26, §§2-4]

2. Relations, symmetry and coherent composition

Finite relational structures give a precise example of representation-dependent reduction. In a fixed positive-arity signature, ordinary homomorphisms define equivalence, and every nonempty finite structure has a core unique up to isomorphism within its equivalence class. Products and disjoint unions induce meet and join on the quotient. Counting, deletion and representative selection need not survive that quotient. Uniqueness of a normal form therefore does not select a physical structure. The empty-relation singleton is weakly initial, not initial, in the full nonempty category; the thin quotient retains a least object. [11, §§2-4; 21, §2]

The review notes report several global-compatibility results under separate hypotheses; the cited summary passages do not supply complete proofs for all of them. Coherent finite nonempty systems over a suitable cofiltered index admit inverse-limit assignments; when there is a greatest context, the limit is its fiber and need not be a singleton. Arbitrary contextual covers do not inherit this result. Proper marginals can conceal parity information, and finite exchangeability need not extend indefinitely. Classical simultaneous assignments, Hilbert-space measurements and unrestricted no-signalling boxes have different Bell bounds because they impose different response structures. [25; 4, §8]

Equality patterns provide a further mathematical laboratory. Relabeling yields set-partition orbits, but symmetry does not choose a probability law. Uniform finite partition rules can fail restriction consistency, while coherent paintbox and Ewens constructions retain choices. Different coherent families can agree through any prescribed finite order. Amalgamation, independent refinement, meet and join have different domains and compatibility properties; separate marginal laws do not determine their coupling or aligned observations. [13, §§2-4; 16, §§2-5; 22, §§2-3]

Strong assumptions nevertheless yield genuine selection results. Deterministic binary operations on finite equivalence relations that are natural under every injection form a five-operation class; commutativity and pointwise idempotence select meet. Stochastic versions require explicit conditional-composition hypotheses. Approximate associativity supplies a size-dependent estimate under exact restriction compatibility, without automatically yielding uniform control over arbitrarily large carriers. [14, §§2-4; 15, §1]

A reported matching-positivity result constrains a loop weight to nonnegative integers without selecting one integer or a physical dimension. Positive correlations likewise need not become a metric under a chosen logarithmic transform. Relative-reference constructions preserve group actions, matching orbits and loop information, but require specified alignment and readout. Formal gauge covariance, cocycles and holonomy do not by themselves identify physical gauge fields or spacetime curvature. [25; 23, §§3-6]

3. Information, signed balance and resource accounting

Signed cancellation can coexist with substantial structure: a zero signed sum, parity trace or waveform integral does not determine a state law or erase all observations. Sector-leakage estimates depend on both escape and entry. States can become invisible to fixed compact observations while remaining distinguishable by moving or richer effects. These are statements about an observation class, not universal annihilation or the derivation of a protected physical sector. [12, §§2-5; 25]

Information bounds require declared preparations, decoding, assistance, composition and costs. A common normalized reference and a bound on the complete preparation’s relative entropy imply a corresponding bound on accessible information. Marginal costs alone miss information encoded in joint correlations. Hard one-mark and mean-one constraints yield different capacities, and both grow with the number of accessible positions. Unlimited independent copies also change the total resource problem. [12, §6; 22, §4]

For a fixed countable alphabet, a nonnegative cost r, and a distribution with mean cost at most E, take $\beta>0$ with finite $Z(\beta)$. With natural logarithms and entropy in nats, Gibbs summability gives the conditional bound

$$H \le \beta E + \log Z(\beta), \qquad Z(\beta) = \sum_j e^{-\beta r(j)} < \infty.$$

The inequality does not choose the costs, their physical units or the accessible alphabet. Finite hard-cost sublevels alone need not bound entropy under a mean-cost constraint. [22, §4]

Partition-output studies further separate entropy from controllable communication. A coherent random output can have conditional entropy unavailable to a sender who controls only its order. Under the stated order-only, mean-one contract allowing order zero, the communication maximum is two bits; the distinct coherent output-entropy optimization remains open. Mathematical bits per use acquire a physical rate only when a use and its timing are operationally defined. [13, §5; 14, §5]

4. What observations identify: operators, spectra and dynamics

A property is identified precisely when it is constant on the fibers of the admitted outcome-law map. Source, detector and calibration parameters must remain variable whenever the experiment permits them to vary. In finite operator algebras, commutant multiplicities control invariant rank-one resolutions; their existence or uniqueness within a supplied algebra does not identify physical subsystems. Compactness and continuity can provide qualitative inverse stability without a numerical error modulus. [2; 27, “Identification from observations”]

Graph experiments illustrate both recovery and blindness. Suitable labeled coherent probes and calibrated scale observations can recover a generator, while spectral or phase-restricted menus admit aliases. Separate finite-graph notes report action-count aliases and recovery from complete labeled sign laws under a specified response model. Positive-measure moment and Laplace constraints instead define feasibility regions and support bounds; numerical candidates and certified dual bounds retain different evidential status. [5; 24, “Finite graph models”; 27, “Spectral identification, detectors and decision rules”]

Fermionic examples separate a static state from its dynamics. Positive rescaling of a gapped Hamiltonian can preserve its occupied projector while changing excitation energies. Clock and generator rescalings can compensate. The calibrated two-time and thermal identification results therefore retain their parameter-domain, detector, normalization and timing assumptions. Compact-group transfer calculations likewise concern a supplied group, kernel and topology; fixed-time convergence does not establish every moving-window or ultraviolet limit. [6; 7; 28]

Continuous-time Markov examples sharpen the timing issue. Distinct generators can give identical full transitions at every integer time, including histories with common interventions confined to that grid. An intermediate calibrated measurement separates the exhibited pair. Conversely, exact full matrix exponentials at two known positive times of irrational ratio identify a common finite-dimensional homogeneous generator. Neither statement derives a physical clock; exact uniqueness also fails to supply uniform stability over unbounded rates. [4, §§6-7]

5. Quantum operations, records and their mathematical assumptions

Conditional instruments are studied with a common completely positive operation, specified tensor factors and complete attempt-level success information. Subnormalized marginals constrain the weight in a selected operator subspace. Binary discarded factors yield sharp intervals; higher-dimensional cases require joint block-Gram feasibility rather than independent pairwise optima. Calibrated coherent return or known coarse flag responses can tighten identification. Phase scans, complete single-input outputs and process tomography remain distinct observation menus: matching restricted records can coexist with different Choi purities or a nonunitary selected branch. [3, §§1, 3-5; 29; 30]

The generalized-event notes report tests of the boundary of the quantum interpretation; their summary passages retain a limited proof status. Positivity with a nonstandard normalization need not assign ordinary probabilities to every event. A specified aggregation expands trace distance and therefore cannot be realized exactly by a deterministic quantum channel on the designated pair. Approximation and heralded constructions have different feasible operations and resource accounts. Conjugation ambiguities, support quotients and alternative decoherent partitions depend on the effects admitted. [25; 1, §7]

The signed-source notes report quantum completions that explicitly add complex qubit algebras, Born effects and tensor products. The available passages mix stated results with selected arguments. A projective separable family is not automatically generated by a copying dynamics; nonorthogonal branches obstruct perfect extension by a branch-blind channel. Identical signed measurement laws can coexist with different transverse coherence. Adjoining a swap operation or noncommuting algebra is a further postulate. Support-restricted states also require care before invoking faithful modular dynamics. [24, “Shared-latent quantum” and selected arguments]

Phase-paired, Bell-resource and collective-spin constructions compare additional choices. Phase gates can preserve signed outcomes while changing entanglement and Bell statistics. Casimir objectives have degeneracies and projectivity restrictions; chosen dimer families and separable exchangeable mixtures provide positive exceptions to overly broad impossibility claims. A matching, objective or reference defined using a desired resource does not independently derive that resource. Entanglement, Bell violation and accessible discrimination are different conclusions. [24, “Phase-paired quantum”, “Collective spin” and §§4.5-4.12]

For identical conditionally independent branch records, a conventional interference trial gives

$$p_0 = \frac{1 + \mathrm{Re}\left(e^{i\phi}\gamma^N\right)}{2}.$$

The forward calculation survives its identification limits. A declared classical response reproduces the restricted real-overlap output curve, and a circle-valued process extends this representation to complex overlap. These constructions concern the specified final outcome, not unrestricted quantum experiments. An environment-only unitary can preserve system channels while changing fragment correlations. History equivalence additionally requires the transformed fragments to remain inaccessible and never recouple; calibrated fragment access supplies a positive discriminator for the displayed pair. [4, §§2-3]

6. Reduced dynamics, memory and hidden realizations

Exact autonomous reduction of a controlled finite Markov model requires equal pushed-forward transition rows within each observational fiber, for every admitted action. This strong lumpability condition applies to all allowed preparations and visible-history policies. A fitted stationary pair matrix is weaker: processes can share single and adjacent-pair laws yet differ at three outputs. Uniform conditional row errors yield complete-history bounds; average errors need not. [17, §§3-4, 6]

Parity-memory examples can hide dependence through a prescribed block length. Their ideal transformed innovations may be independent, whereas overlapping innovations after noisy readout are generally dependent. Attenuating a mean is therefore insufficient to preserve a binomial power calculation. Exact realization rank gives classical state-size lower bounds only within the assumed across-cut factorization; it is neither a count of physical constituents nor automatically a quantum-memory bound. [18, §§2-6; 21, §5]

Even minimal positive hidden models can share complete word laws while differing in numerical parameters. Calibrated controls can break that ambiguity: a determinant condition characterizes the stated two-control, two-state reconstruction. Allowing candidate-dependent control uncertainty restores an explicit compatible interval, while exact reset-and-cycle controls give a separate finite-state positive result. These conclusions depend on anchoring, linear observability and the admitted control class. Population uniqueness, finite-sample stability and calibration cost remain different questions. [19, §§6-7; 4, §§4-5]

7. Statistical evidence, comparisons and calibration

Sequential instrument tests retain every attempt, including failures, and fix eligibility before private assignment. Under the declared conditional null, suitable nonnegative products control the probability of ever crossing a threshold. Success-only selection can invalidate that guarantee. Reported rare-success sample counts distinguish necessary bounds, attained simple-law tests and composite alternatives; they do not automatically become device operating budgets. [3, §6; 27, “Sequential evidence and selection”]

Full-record likelihood mixtures extend detection to unknown memory lengths under a specified observed null. Eventual detection for each fixed alternative does not yield a uniform finite horizon over arbitrary lengths or near-erasure readout. A factor may remain valid while having poor growth; growth depends on how the alternative lies outside the conditional null envelope. Widening that envelope can include the proposed alternative itself. [21, §§3-4]

Collision and equality-comparison statistics require their actual dependence structure. Overlapping subsets are not independent. Corrupting raw labels, corrupting pair reports and applying transitive closure produce different sensitivity bounds. Sparse random querying saves comparisons under a committed report population and an independent query seed; query-aware corruption can defeat that contract. Label count, query count, power and robustness are separate resources. [15, §§3-5; 20, §§2-4]

Calibration audits also separate coverage from acceptance and power. A quality gap can permit useful finite sampling; distinguishing adjacent worst-case error counts at a sharp threshold may

require inspecting almost the entire finite population. The latter optimum concerns truthful audit-only access without informative side channels. Agreement between two references does not establish either reference's correctness: a shared coarse error can erase the distinction of interest. Physical reference accuracy and future drift need additional evidence. [19, §§2-5; 20, §§5-6]

8. The analytic physics program and its remaining correspondences

The Wilson-Dirac work studies interval restrictions of infinite-chain occupied-band quasifree states, with bare mass $m=\mu/L$. The original relative-entropy manuscript states a leading fixed-positive-μ logarithmic law. The later *Finite Part of Wilson-Dirac Relative Entropy* states its constant term and inverse-logarithmic remainder. These are distinct analytic assertions with specified reference geometry and concentration, Jacobi, majorant, nonlinear and boundary inputs. Their overview does not substitute for checking the full dependency chain. [8; 9]

Uniform Modular Control and Nonlinear Convergence for Wilson-Dirac Intervals supplies a uniform modular norm bound and sharper weighted errors. Fixed-index, bulk and moving-window estimates retain different ranges. Compression requires particular care: an operator lying between two signed bounds need not satisfy the corresponding absolute-value bound. Positive-majorant arguments cannot be replaced by that invalid inference. [10; 8, Appendix C]

Boundary trace-class and covariance estimates include a positive uniform entropy-subsystem result over bounded mass intervals containing zero. This does not establish uniformity of the entire finite-part formula in a shrinking-mass regime. Improvements to modular or nonlinear errors also do not automatically remove the boundary bottleneck. [8, Appendix C; 9; 10]

Other conventional comparisons retain their own mathematical and physical assumptions. Proposed field, classical-motion, horizon and gravitational interpretations require the specified states, dynamics, metrics and normalization choices. The program's next physical claims must specify how mathematical states, transformations and quantities correspond to preparations, controls and readouts, and what calibrated observations discriminate the resulting models. A warranted partial correspondence would be meaningful progress while broader questions about clocks, geometry, matter and a unified physical account remain open. [26, §§2-5; 11, §§7-10]

References

The research notes identify Kirtus Justus as author or research director and disclose mathematical or drafting assistance where applicable. These are sources within the program, not independent replications.

1. Relational Opposition, Gauge Redundancy, and Time-Unassigned Information. Research note, 8 November 2026, §§2-8.

2. Invariant Relational Selection and Approximate Symmetry. Kirtus Justus, 5 November 2026.

3. Identification from Restricted Measurements. Kirtus Justus, expanded mathematical manuscript, §§1-8.

4. Operational Identification of Records, Controls and Continuous-Time Dynamics. Mathematical study, §§2-8.

5. Positive Spectral Measures Under Finite Moment and Laplace Constraints. Kirtus Justus, 23 October 2026.

6. Operational Identification and Calibration in Two-Mode Quasifree Systems. Kirtus Justus, 23 October 2026.

7. Compact-Group Transfer Spectra and Ultraviolet Operator Limits. Kirtus Justus, 23 October 2026.

8. Relative Entropy of Finite-Interval Wilson-Dirac Quasifree States. Kirtus Justus, 31 October 2026; including the specified boundary-entropy supplement.

9. Finite Part of Wilson-Dirac Relative Entropy. Kirtus Justus, 31 October 2026.

10. Uniform Modular Control and Nonlinear Convergence for Wilson-Dirac Intervals. Kirtus Justus, 31 October 2026.

11. Relational Reduction and Conditional Physical Completion. Kirtus Justus, 9 November 2026.

12. Observation Contracts, Repetition Resources, and Sharp Sector-Leakage Bounds. Research note, 9 November 2026.

13. Finite-Order Observation, Compositional Typing, and Coherent Entropy in Exchangeable Partitions. Kirtus Justus, 9 November 2026.

14. Restriction-Compatible Stochastic Operations and Information Accounting. Kirtus Justus, 9 November 2026.

15. Local Stochastic Stability and Comparison-Readout Robustness. Kirtus Justus, 9 November 2026.

16. Coupling, Joint Observation, and Moving-Arity Identification in Exchangeable Partitions. Research note, 9 November 2026.

17. Reduced-State Sufficiency and Finite-Memory Controls. Research note, 9 November 2026.

18. Noisy Predictive Memory and Exact Detection. Kirtus Justus, 9 November 2026.

19. Sharp Audit Boundaries and Minimal Stochastic Realizations. Kirtus Justus, 9 November 2026.

20. Sampling and Calibration of Robust Equality Comparisons. Kirtus Justus, 9 November 2026.

21. Categorical Corrections and Sequential Evidence. Kirtus Justus, 9 November 2026.

22. Invariant Equality Patterns, Probability Lifts, and Resource-Growth Bounds. Research note, 9 November 2026.

23. Gauge-Orbit Composition and Operational Equivalence. Research note, 8 November 2026.

24. Unpublished mathematical notes, 8 November 2026: “Finite graph models”, “Shared-latent quantum”, “Phase-paired quantum”, “Collective spin”, and selected arguments in §4. Statements and selected proofs; not a complete proof source for every assertion.

25. Unpublished mathematical review-note passages, 9 November 2026: “Generalized event kernels are not automatically quantum channels”; “Marginals, exchangeability, and contexts”;

“Spectra, geometry, balance, and observation limits”. Source-reported assertions; full proofs are not contained in these passages.

26. From Prephysical Hypotheses to Physical Models. Kirtus Justus, 31 October 2026, individual introductory article, §§1-5.

27. Operational Identification and Conditional Quantum Instruments. Kirtus Justus, mathematical manuscript.

28. Projector-Equivalent Wilson-Dirac Hamiltonians and Dynamical Identification. Kirtus Justus, 23 October 2026.

29. Joint Purification Constraints on Local Kraus Weight: Exact Attainment, Simplex Obstructions, and Error Stability. Research note, 7 November 2026.

30. Flag-Resolved Identification of Local Choi Weight: Joint Semidefinite Bounds, an Exact Three-Level Example, and a Gibbs-State Check. Research note, 7 November 2026.